GLGAT-CFSL: Global–Local Graph Attention Network-Based Cross-Domain Few-Shot Learning for Hyperspectral Image Classification

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***Abstract*— Few-shot learning (FSL) is an effective approach to address the issue of limited labeled data in hyperspectral image classification (HSIC). However, it overlooks the domain shift between the source domain (SD) and the target domain (TD) in cross-domain tasks. Most existing domain adaptation (DA) methods alleviate the domain shift problem to some extent, but DA methods based on traditional convolutional operators overlook the nonlocal spatial relationships in HSI, while methods based on graph neural networks (GNNs), although effective in leveraging nonlocal spatial information for domain alignment, overly emphasize global relationships, which is disadvantageous for pixel-level classification in HSI. To solve these issues, this article proposes a novel globalp–local graph attention network-based cross-domain FSL (GLGAT-CFSL), which comprehensively reduces domain shift through global-to-local domain alignment. It has the following advantages: 1) an innovative dynamic triplet graph attention network is devised to identify nonlocal spatial relationships in HSI for global graph alignment (GGA) while also addressing common overfitting and oversmoothing issues in GNNs; 2) an ingenious local similarity learning (LSL) strategy is designed after global domain alignment, utilizing intradomain connectivity structures and interdomain node similarities for local DA, promoting cross-domain information propagation and more comprehensive reduction of domain shift; and 3) we propose a novel triaxial dynamic convolutional neural network (TDCNN) as the feature extractor, promoting cross-dimensional interaction between spectral and spatial dimensions, establishing a more generalizable and rich feature representation between the SD and the TD. The experimental results on three HSI datasets demonstrate the superiority and effectiveness of the proposed GLGAT-CFSL.**

***Index Terms*— Cross-domain hyperspectral image classification (HSIC), domain adaptation (DA), domain shift, few-shot learning (FSL), graph neural networks (GNNs).**

I. Introduction

REMOTE sensing images are characterized by high resolution, all-weather coverage, and wide-area observation, with applications in agriculture, environmental monitoring, and geological exploration being of significant importance [1]. Hyperspectral imaging is a hot research topic in the field of remote sensing, involving tasks such as classification, unmixing, and super-resolution [2], [3], [4]. Hyperspectral image classification (HSIC), in particular, entails utilizing spectral and spatial information to determine the category of individual pixels, thereby enhancing understanding of the surrounding environment. Therefore, the comprehensive utilization and integration of spectral and spatial data hold significant research implications for improving the accuracy of HSIC [5]. Hong et al. [6] proposed a novel 3-D generative pretrained transformer (GPT) to address the challenge of existing models in capturing spectral information in remote sensing image data processing, enhancing the understanding and analytical capabilities of spectral data through progressive training and spatial–spectral coupling. However, achieving high accuracy in HSIC remains challenging because of the complex nature of the high-dimensional spectral signal and the restricted number of labeled training samples [7], [8], [9].

In order to address these issues, numerous learning strategies have been extensively studied [10], [11], [12], [13]. Semi-supervised learning [14], [15], [16] and unsupervised learning [17], [18] have been employed to alleviate the problem of insufficient labeled data. While these methods excel in addressing the challenge of limited labeled data, they may struggle to distinguish these new classes when faced with samples from classes not encountered during the training phase due to a lack of sufficient prior knowledge. In contrast, inspired by the human visual perception system, few-shot learning (FSL) rapidly leverages limited annotated data to acquire prior knowledge, successfully generalizing it to the classes unseen during the training phase, making it to emerge

Manuscript received 27 March 2024; revised 28 April 2024; accepted 21 May 2024. Date of publication 14 June 2024; date of current version 16 July 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 61901369, Grant 62071387, and Grant 62101454; and in part by the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing Foundation under Grant TJ-04-23-04. *(Corresponding author: Yanning Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2024.3407812

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as one of the most effective learning paradigms [19], [20]. Through the utilization of meta-learning, many FSL methods based on metric and embedding learning methods have been applied to HSIC tasks [8], [21]. Liu et al. [22] proposed a deep FSL (DFSL) method, which leverages the Euclidean distance within a metric space to distinguish different classes and employs a nearest neighbor algorithm for classification, addressing the challenge of decreased model classification performance in HSIC due to the scarcity of labeled samples. Li et al. [23] designed an RS-MetaNet to learn a task-level distribution through task-based metrics, improving its generalization to unseen test tasks. Zhang et al. [24] proposed a global prototypical network (GPN), which learned the global prototype representation of each class in the embedded feature space through a global representation learning strategy, enhancing the model's generalization ability in a few sample scenarios.

The FSL methods mentioned above can improve the classification accuracy for HSI with a small number of labeled samples, but they typically assume that both the training and testing data originate from domains with the same distribution [25], [26]. However, in practice, the data distribution of the source domain (SD) and the target domain (TD) may be inconsistent, which necessitates the model trained on the SD to handle new classes and domains, leading to a decrease in the model's generalization ability, namely, domain shift [27]. Domain adaptation (DA) strategies have been extensively employed to address the challenges of cross-domain learning tasks [28]. There are three main types of DA methods: adversarial-based, discrepancy-based, and generative-based [27], [29], [30]. Xi et al. [31] proposed a cross-domain FSL framework for HSIC. The framework utilized class-covariance measurement to learn global class representations and introduced a spectral-prior-based refinement module to focus on the most informative spectral bands, addressing the impact of mitigating domain shift. Ye et al. [32] achieved global conditional distribution alignment using adaptive conditional adversarial losses and combined it with a multiscale spectral–spatial feature extraction (MSFE) module to enhance cross-domain FSL, alleviating the time-consuming issue of annotating hyperspectral image (HSI) data. Huang et al. [33] proposed a novel cross-domain FSL framework that addresses the challenge of limited labeled training samples in HSIC. By considering all possible positive and negative sample pairs in a training batch and introducing a new kernel triplet loss to characterize complex nonlinear relationships between samples, they aim to resolve this challenge.

DA methods have alleviated the domain shift problem to some extent in HSIC. However, DA methods based on traditional convolutional operators overlook the nonlocal spatial relationships in HSI, and relying solely on traditional local information may not effectively address the domain shift problem [34]. DA methods based on graph neural networks (GNNs) effectively capture and utilize nonlocal spatial information in HSI through their graph information aggregation characteristics [34], exhibiting strong performance in cross-domain few-shot classification tasks. However, they overly emphasize global relationships at the expense of individual pixel features, which is disadvantageous for HSI that requires accurate classification of each pixel [35], [36], [37]. In addition, there exist complex nonlinear relationships between spectral and spatial dimensions in HSI, and most existing feature extraction methods fail to capture the interaction information between spectral and spatial dimensions in HSI, resulting in inadequate feature representations [38], [39].

In order to solve the aforementioned issues, we propose an innovative global–local graph attention network-based cross-domain FSL (GLGAT-CFSL) framework. We have designed two novel and effective domain alignment methods, namely, global graph alignment (GGA) and local similarity learning (LSL), which complement each other to more comprehensively represent domain distributions. First, GGA utilizes an ingenious dynamic triplet graph attention network, comprising node self-attention, graph neighbor attention, and memory-layer attention, to generate more discriminative features for global domain alignment while preventing model overfitting and oversmoothing. Second, LSL conducts node-line joint matching by simultaneously considering node and line information, fully leveraging the intradomain connectivity structure and interdomain node similarities in HSI for local DA, promoting cross-domain information propagation and node alignment, and further reducing domain shift. Third, we propose a novel triaxial dynamic convolutional neural network (TDCNN) as the feature extractor. It employs triaxial attention (TA) with three parallel branches to facilitate cross-dimensional interaction between spectral and spatial dimensions, thereby establishing a more generalizable and rich feature representation between the SD and the TD while also constructing more accurate graph structure information. Finally, GLGAT-CFSL combines DA based on GLGAT with FSL, executing FSL on both SD and TD simultaneously, enabling the discovery of transferable knowledge from the source classes and learning a discriminative embedding model for the target classes, thereby enhancing the network's performance and generalization in the TD.

The primary contributions of this article are summarized as follows.

1) A novel framework GLGAT-CFSL, which combines DA based on GLGAT with FSL, has been proposed. It enables the model to generalize on the TD by leveraging limited labeled samples from the SD and effectively conducts domain alignment using global–local domain alignment, achieving accurate cross-domain HSIC.
2) An innovative GGA method has been devised, which conducts global DA through an ingenious dynamic triplet graph attention network. It performs GGA by leveraging sample correlations at both the feature and class levels, generating more discriminative features. Furthermore, it addresses common issues in GNN, such as overfitting and oversmoothing, by incorporating sparsity constraints and memory attention.
3) An innovative LSL strategy has been devised, which complements GGA. It simultaneously considers both node and line information in HSI, facilitating local domain alignment through node-line joint matching. By leveraging the connectivity structure within images,

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it promotes cross-domain local information propagation, further mitigating the negative impact of feature mismatch caused by domain shift.

4) A novel feature extractor, TDCNN, is proposed to establish a more consistent and comprehensive feature representation and to construct more accurate graph structure information. TDCNN promotes cross-dimensional interaction between spectral and spatial dimensions through TA with three parallel branches and enhances model expressiveness by integrating multiple parallel convolutional kernels through dynamic convolution.

The subsequent sections of this article are structured as follows. Section II presents an overview of the key concepts of GNN, cross-domain FSL, and DA. In Section III, a comprehensive introduction to the GLGAT-CFSL method is provided. The ablation and comparative experiments of GLGAT-CFSL outcomes concerning three publicly available HSI datasets are analyzed in Section IV. The discussion on hyperparameters and TDCNN is in Section V. The conclusions are presented in Section VI.

II. Related Work

This section introduces some works related to GNNs, cross-domain FSL, and DA.

A. Graph Neural Networks

GNNs represent the relationships between samples as data with a graph structure, aiming to perform representation learning on nodes within graphs for various tasks [40]. Graph data consist of nodes and edges connecting them. The core idea is to update node representations by iteratively aggregating neighbor information [41]. The iterative aggregation process enables nodes to capture a broader context, enhancing the expressive power of node representations [42]. Therefore, GNNs are also well-suited for HSIC. Hong et al. [43] addressed the computational cost and inference challenge of traditional graph convolutional networks (GCNs) in large-scale remote sensing problems by proposing miniGCN, which trains large-scale GCNs through mini-batch processing to reduce computational costs, achieving superior performance in HSIC tasks. The typical GNN framework can be formulated as

$$H_v^m = \text{Combine}\left(H_v^{(m-1)}, \text{Aggregator}\left(H_u^{(m-1)}\right)\right) \quad (1)$$

where H_v represents the vector representation of the central node and H_u represents the vector representation of the central node. The combine function and aggregator function are utilized to update the node's representation. After mm rounds of aggregating information for each node v, a new representation vector H_v^m is generated, which is then input into a mapping function to obtain the final results for a specific task.

Recently, GNNs have shown potential in cross-domain few-shot HSIC. Wang et al. [44] embedded domain-wise correlation alignment and class-wise correlation alignment in GNN to address the DA issue in multitemporal HSIs. Zuo et al. [45] proposed an FSL method based on edge-labeling GNN (FSL-EGNN) to explicitly quantify intraclass and interclass associations between pixels, addressing the challenge of few sample HSIC and outperforming existing methods on benchmark datasets. Although GNNs perform strongly in cross-domain HSIC, leveraging their effective ability to capture nonlocal spatial relations, most GNN-based DA methods encounter issues such as overfitting and oversmoothing. In addition, they tend to overlook the complementary relationships between nodes and line segments. In this article, we integrate GNNs with attention mechanisms, employing graph attention networks to address the aforementioned issues.

B. Cross-Domain FSL

The task of cross-domain FSL involves learning between an SD (denoted as D_s) and a TD (denoted as D_t), where the data in the SD and TD obey the joint distributions P_s and P_t, respectively [46]. The class label sets of these two domains are Y_s and Y_t, respectively, and Y_s and Y_t are disjoints.

For the HSI dataset, we divide it into two parts, i.e., the source dataset with a large number of labeled samples and the target dataset with only a small number of labeled samples. In this context, we introduce the SD D_s and the TD D_t, whose distributions correspond to P_s and P_t, respectively. It should be emphasized that P_s is not equal to P_t due to the existence of domain transitions in the SD and TD. In the specific setup, we assume that the numbers of classes and labeled samples in the SD D_s are C_s and N_s, while the numbers of classes and labeled samples in the TD D_t are C_t and N_t, respectively. To ensure diversity in the training samples, C_s is generally made larger than C_t. This ensures that the knowledge learned by the model in the SD can be migrated to the TD and effective classification is achieved in the TD.

Overall, cross-domain FSL aims at solving the problem of domain transformation between the SD and the TD, which involves differences in the joint distribution PP, the set of class labels YY, and the number of labeled samples in different domains. Through methods such as meta-learning, the model can adapt to the TD with a limited number of labeled samples for efficient learning and generalization.

C. Domain Adaptation

Cross-domain FSL focuses on solving classification problems in situations where labeled samples are scarce. It transfers knowledge between the SD and the TD using techniques such as transfer learning or meta-learning. On the other hand, DA aims to address the issue of mismatched data distributions between the SD and the TD by adjusting the model or features to adapt to the TD's data distribution [47], [48]. It improves the generalization ability and practical application performance of the model by adapting the model or features to perform better on the TD [49]. Traditional shallow DA methods are mainly divided into instance- and feature-based approaches [50], [51]. However, their representation learning capability is limited, thus lacking the ability to handle data with complex structures [12], [52], [53]. Compared to shallow DA methods, deep DA methods leverage the automatic learning capability of deep neural networks to acquire more complex and higher level feature representations. As a result, they have gained more attention and have been widely applied

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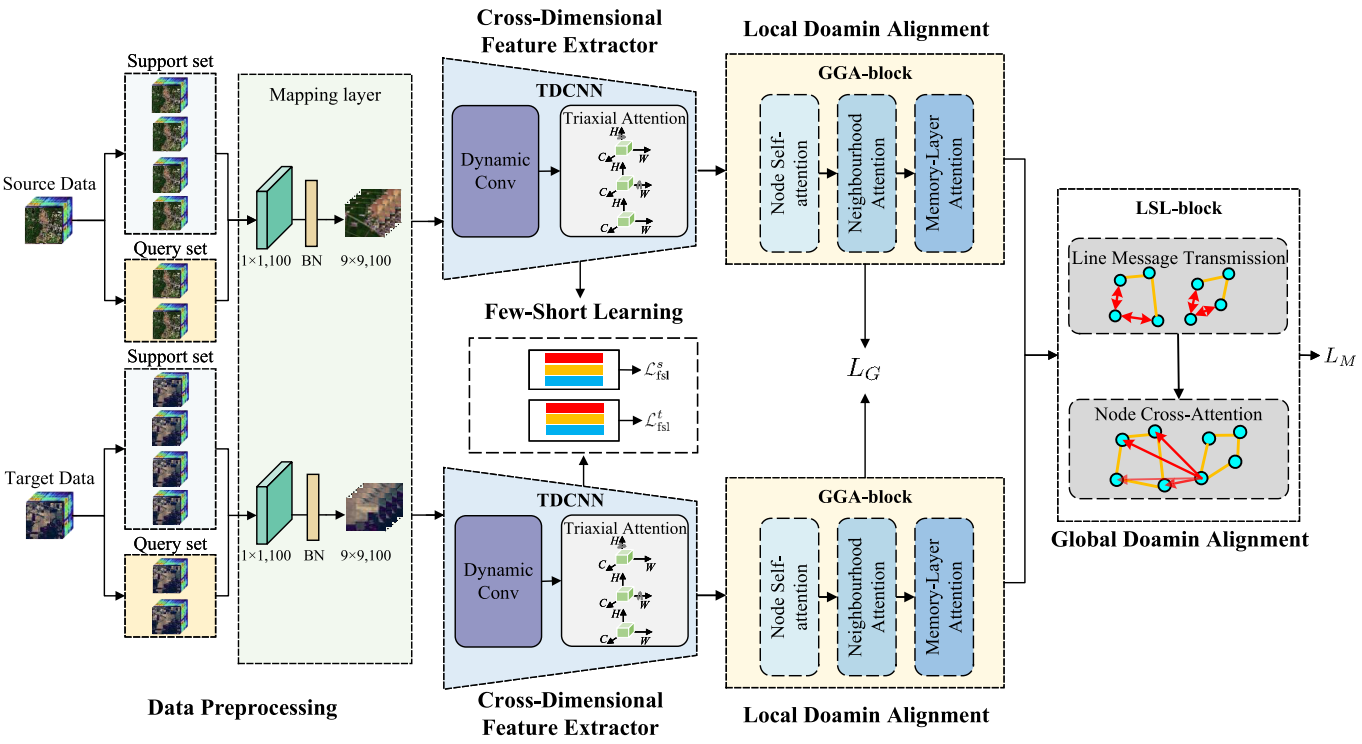

Fig. 1. Diagram outlines the proposed GLGAT-CFSL, including data preprocessing, cross-dimensional feature extraction, FSL, and global–local domain alignment.

in the field of cross-domain few-shot HSIC [29], [54], [55]. Deep DA methods commonly used can be divided into three classes: discrepancy-based, adversarial-based, and generative-based [56], [57], [58], [59]. Compared to adversarial-based methods, discrepancy-based methods may exhibit better convergence and stability, with reduced risk of model collapse. In contrast to generative-based approaches, discrepancy-based methods directly minimize the feature discrepancies between the SD and the TD, avoiding potential biases and distortions introduced during the sample generation process. This helps better preserve the original characteristics and distribution information of the data. In addition, discrepancy-based methods directly minimize the feature differences between the SD and the TD, without introducing additional generative networks or loss functions. Consequently, they are better able to preserve the original characteristics and distribution information of the data, helping to retain the intrinsic features of the data. Therefore, the two DA methods we use in this article, namely, GGA and LSL, are both difference-based methods. They leverage the maximum mean discrepancy (MMD) distance and the graph optimal transport (GOT) distance to collaboratively achieve domain alignment.

III. Methods

During the training phase, as illustrated in Fig. 1, the GLGAT-CFSL framework comprises data preprocessing, cross-dimensional feature extraction, FSL, and global–local domain alignment. First, in the data preprocessing stage, we divide the SD and TD data into support sets and query sets and transform them to the same dimension using mapping layers. Second, for cross-dimensional feature extraction, we employ a novative TDCNN to extract more discriminative spatial–spectral embedded features, facilitating cross-dimensional interaction between spectral and spatial dimensions and enhancing the network's capability to capture key features. Then, we conduct FSL simultaneously on both the SD and the TD using Euclidean metric to facilitate knowledge transfer in cross-domain tasks. Finally, to address the negative impact of domain shift on classification performance during the FSL phase, we introduce GGA and LSL based on graph attention networks, aiming to comprehensively consider both the consistency of overall data distribution and the similarity of local data relationships, thereby more comprehensively reducing domain shift.

During the testing phase, the classification of unlabeled samples in the TD is performed, as illustrated in Fig. 2. First, the dimensionality of input samples is reduced through the mapping layer. Next, spatial–spectral embedding features are extracted using the TDCNN feature extractor. Finally, a K-nearest neighbor (K-NN) classifier is employed to classify the unlabeled samples, ultimately yielding the classification results for the TD.

A. Data Preprocessing

As shown in Fig. 1, we commonly represent the input HSIs from the SD and the TD as $R^{H\times W\times \text{ch1}}$ and $R^{H\times W\times \text{ch2}}$, respectively, where H and W denote the height and width of the HSI, respectively, and ch1 and ch2 represent the HSI's spectral dimension in the SD and TD, respectively. First, taking the SD as an example, we extract $9 \times 9 \times$ ch1 neighborhood data cubes centered at each pixel from $R^{H\times W\times \text{ch1}}$. The same operation is performed for the TD. Second, both the SD and TD's HSI data are divided into support set S and query set Q to simulate the scenarios of FSL

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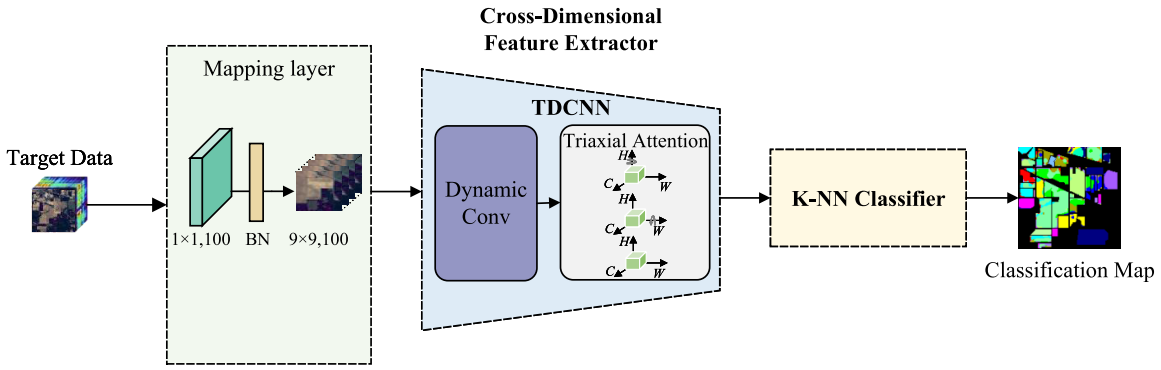

Fig. 2. Diagram illustrates the testing phase workflow of GLGAT-CFSL. It involves extracting features from the TD using the trained TDCNN and classifying unlabeled samples using a K-NN classifier.

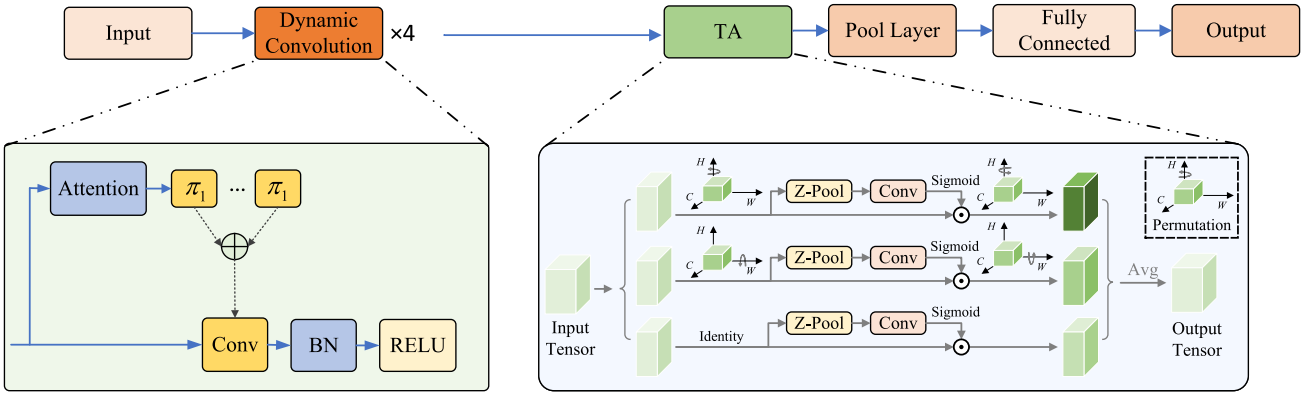

Fig. 3. Structure diagram of TDCNN. Input image undergoes dynamic convolution to generate output features. Subsequently, three branches capture interactions between channel dimensions and spatial dimensions, and process features through Z-Pooling and convolutional layers. Finally, the output features from the three branches are summed and averaged to obtain the final feature representation.

in real-world applications. The support set is a dataset used for training the model, while the query set is a dataset used for evaluating the model's performance. Finally, to address the issue of inconsistent spectral resolution among samples due to different sensors in the SD and TD, a mapping layer is employed to align the HSI features from the SD and TD to the same dimension. After inputting the HSI data cubes of size $9 \times 9 \times$ ch1 and $9 \times 9 \times$ ch2 from both the SD and the TD into the mapping layer, the spectral dimensions of both the SD and the TD are transformed to a $9 \times 9 \times 100$ dimensional space. It ensures a consistent comparison between the support set and the query set within the same dimensional context, with 9×9 representing spatial dimensions and 100 denoting the spectral dimension.

B. Cross-Dimensional Feature Extraction

The SD and the TD may have different feature distributions and data representations in cross-domain HSIC tasks. Through cross-dimensional interaction, information from different dimensions can be integrated to better capture shared features between domains, thereby enhancing the model's performance and generalization capability on the TD. Moreover, in HSI, the spatial distribution of objects is closely tied to their spectral characteristics, making cross-dimensional interaction crucial for accurately representing the graph structure. Therefore, in order to facilitate cross-dimensional interaction between spectral and spatial dimensions in HSI, we propose a novel TDCNN as a feature extractor.

As shown in Fig. 3, the TDCNN adopts the residual network as its fundamental framework, combined with dynamic convolution and TA. Dynamic convolution enhances the model's expressive capability by integrating multiple parallel convolutional kernels, and its process can be succinctly expressed as

$$\hat{y} = g\left(\hat{W}^T \hat{x} + b\right) \tag{2}$$

where $\hat{W}$, b, and g, respectively, represent the attention weight, bias, and activation function, with the weights changing in response to variations in the input data. The HSI data cube $\hat{x}$ with preprocessed dimensions of $9 \times 9 \times 100$ undergoes dynamic convolution to produce the output $\hat{y}$.

In addition, in order to integrate information from different dimensions and better capture shared features between domains, TA is introduced after the dynamic convolutional layers. For a given input of size $W \times H \times C$, the specific implementation process of TA is given as follows.

Step 1: The first branch captures the interaction between channel C and spatial dimension H. The input features undergo cross-dimensional interaction and transform into features of dimensions $C \times H \times W$. Z-Pooling is applied along the W dimension, where Z-Pooling includes both max pooling and average pooling. The output is features of dimensions

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TABLE I
NETWORK LAYER SIZES FOR THE FEATURE EXTRACTION

Layer Name	Output Size	Filter Size	Padding	Stride
Input	9x9x100	N/A	N	N
Dynamic Conv1	9x9x60	3x3	Y	1
Dynamic Conv2	9x9x120	3x3	Y	1
Dynamic Conv3	7x7x120	3x3	N	1
Dynamic Conv4	5x5x64	3x3	N	1
Triaxial Attention	5x5x64	N/A	N	N
AdaptiveAvgPool	1x1x64	N/A	N	N
Linear	64	N/A	N	N

$C \times H \times 2$. Afterward, the features go through a 7×7 convolutional layer and a batch normalization (BN) layer. Through cross-dimensional interaction, the features are transformed into dimensions $W \times H \times C$.

Step 2: The second branch captures the interaction between channel C and spatial dimension W. The input features first undergo cross-dimensional interaction and transform into features of dimensions $H \times W \times C$. Subsequently, Z-Pooling is applied along the H dimension. Then, the features go through a 7×7 convolutional layer and a BN layer. Finally, through cross-dimensional interaction, the features are transformed into dimensions $W \times H \times C$.

Step 3: The third branch is the spatial attention computation branch. The input features go through channel pooling, a 7×7 convolutional layer, and a BN layer. Finally, a sigmoid activation function is applied to generate spatial attention weights.

Step 4: Add the output features from the three branches and take their average.

TA directs the network's attention to crucial areas, extracting more valuable spatial and spectral information. Simultaneously, through cross-dimensional interaction, it integrates information from different dimensions, better capturing shared features between domains, thereby enhancing the model's performance and generalization ability on the TD. In addition, TDCNN constructs a more accurate graph structure information through cross-dimensional interaction, facilitating subsequent domain alignment based on graph attention networks. Table I presents a detailed summary of the layer types and the sizes of input and output feature maps for TDCNN. Finally, the extracted cross-dimensional interaction features are classified using a fully connected layer based on the softmax function.

C. Few-Shot Learning

In cross-domain HSIC tasks, the TD may have very few labeled samples available. By performing FSL in both the SD and the TD, the model can better transfer knowledge learned in the SD to the TD, which helps improve the model's generalization ability to new classes, enabling it to better handle unknown classes that may appear in the TD. Therefore, we conduct meta-learning-based FSL in both the SD and the TD after feature extraction.

During the FSL training process in the SD, N classes are randomly selected from the N_S source classes to form episodes. In each episode, FSL is performed by computing the embedded features of the query set with the prototype of each class using Euclidean distance to perform FSL and minimize the prediction loss of the query set. The predictive probability of the query sample x_i is determined by measuring the Euclidean distance and applying the softmax function. The prediction process can be defined as

$$P(y_i = k \mid x_i \in Q_s) = \text{Softmax}\left(-G_{\text{ED}}\left(c_i^s, c_k\right)\right) \tag{3}$$

where the embedding features of the query set are denoted as c_i^s, indicating the centroid or representative feature of the class in the feature space. c_k denotes the embedding feature associated with the kth class of samples in the support set, indicating the position of the class in the feature space. G_{ED} represents the Euclidean distance function, used to calculate the distance between two feature vectors, ensuring that in this space, similar data points are closer together, while different data points are farther apart. x_i is a sample in the query set, i.e., the sample to be classified. y_i is the label of sample x_i in the query set, i.e., the true class.

Based on the predicted probability outcome for the query sample, the FSL loss for the query sample in the SD can be represented by the following equation:

$$\mathcal{L}_{\text{fsl}}^s = \frac{\sum_{(x,y\in Q_s)}(-\log P(y_i = k \mid x_i \in Q_s))}{|S_s| \times |Q_s|} \tag{4}$$

where S_s represents the size of the support set in the SD and Q_s represents the size of the query set in the SD.

Similarly, the FSL loss for the query samples in the TD can be expressed as

$$\mathcal{L}_{\text{fsl}}^t = \frac{\sum_{(x,y\in Q_t)}(-\log P(y_i = k \mid x_i \in Q_t))}{|S_t| \times |Q_t|}. \tag{5}$$

The total FSL loss can be represented as

$$\mathcal{L}_{\text{fsl}}^{\text{tot}} = \mathcal{L}_{\text{fsl}}^s + \mathcal{L}_{\text{fsl}}^t. \tag{6}$$

D. Global Domain Alignment With GGA

When there are limited labeled samples available in the TD, performing FSL in both SD and TD enables the model to better leverage known knowledge from the SD and transfer it to the TD, thereby enhancing the model's performance in the TD. However, in practice, the data distributions of the SD and TD may be inconsistent, leading to domain shift, which poses challenges for knowledge transfer from the SD to the TD. DA aims to adjust the model to better adapt to the characteristics of the TD. Therefore, this article proposes two efficient graph attention-based domain alignment methods, namely, GGA and LSL, which, respectively, tackle DA from global and local perspectives to achieve comprehensive adaptation to the domain. By combining FSL and DA methods, GLGAT-CFSL overcomes domain shift while effectively utilizing a small number of TD samples, thereby improving the model's generalization performance in the TD.

The design objective of GGA is to achieve global DA by constructing a flexible graph model while mitigating the

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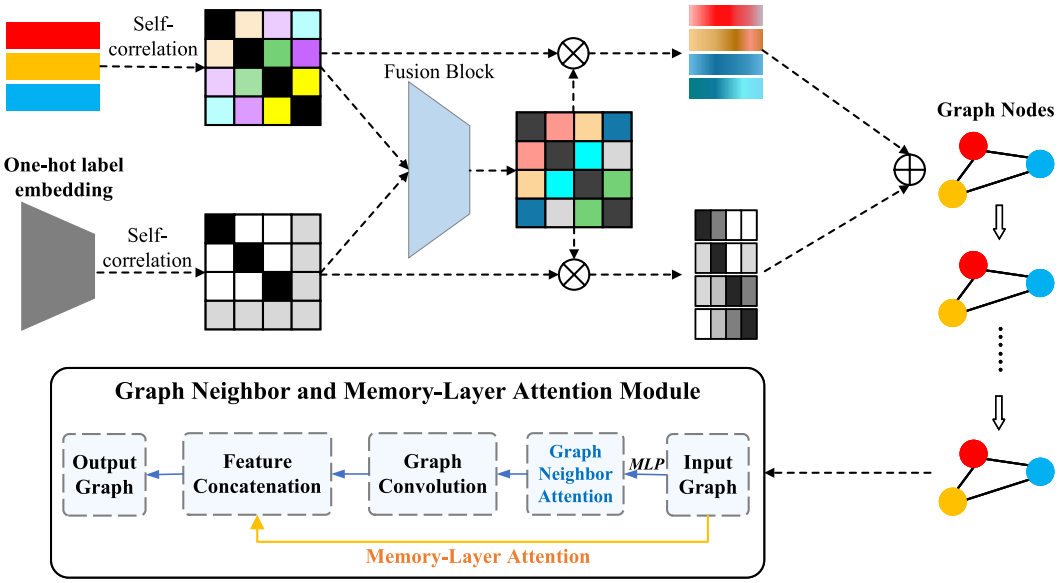

Fig. 4. Schematic of GGA. GGA integrates node features with label information to form initial node features and then dynamically adjusts the graph structure through neighbor attention and memory-layer attention mechanisms, simultaneously addressing the issues of smoothing and overfitting in deep GNNs. Finally, by using the global GOT distance loss, it comprehensively considers both the structure and features of the entire graph, achieving GGA.

common issues of oversmoothing and overfitting in the layers of deep GNNs. As illustrated in Fig. 4, the dynamic triplet attention in GGA includes node self-attention, graph neighbor attention, and memory-layer attention. Its input is the output of the feature extractor, namely, the spatial–spectral embedding features.

1) Graph Initialization: First, node feature extraction and encoding are performed. The TDCNN feature extractor is utilized to extract features from the HSI cubes of both SD and TD, serving as nodes. Simultaneously, in order to better utilize the label information of nodes, the labels of nodes are one-hot encoded, transforming discrete labels into continuous vector representations, aiding the model in more easily learning differences between classes. The vector matrix H can be expressed as

$$H = [h_1, h_2, \ldots, h_i] \tag{7}$$

where h_i represents the one-hot vector of the corresponding label for each node. Next is the computation of node self-attention, obtaining the node self-attention matrix A^x, whose calculation process is as follows:

$$A^x = \text{Softmax}\left[CC^T \odot \left(\|c_i\|_2 \|c_j\|_2\right)^{-1}\right] \tag{8}$$

where $\odot$ represents the element-wise multiplication, c_i denotes the features of each node, and C represents the node feature matrix. Then, label correlation calculation is performed to obtain the label correlation matrix, with the calculation process as follows:

$$A^h = \text{Softmax}\left(HH^T\right). \tag{9}$$

Finally, the feature information is fused with the label information, serving as the initial node features of the graph for subsequent GNN training. The fusion process can be represented as

$$A^f = \text{Fusion}\left(\left[A^h, A^x\right]\right) \tag{10}$$

where $[A^h, A^x]$ represents the weighted sum of the two attention maps. These initial node features contain both the original feature information of the nodes and the label information related to the nodes, providing a foundation for subsequent graph updates and alignments.

2) Graph Updating: Similar to other GNN frameworks, GGA updates nodes by learning the adjacency matrix. However, as the model deepens, GNN tends to blend information from all neighboring nodes, causing node features to converge to the mean and resulting in oversmoothing issues. Second, due to the limited number of samples and the complexity of the GNN structure, overfitting issues may also arise. Therefore, this article shows dynamic sparsity constraints and memory attention mechanisms, designing neighbor attention and layer-memory attention, to address these issues.

Neighbor attention selectively focuses on neighboring nodes relevant to the current node by introducing dynamic sparsity constraints to avoid oversmoothing. Specifically, first, node connection weights F^e are calculated based on the Euclidean distance, which can be represented as

$$F^e(i, j) = \text{PST}\left[G_e d^2\left(c_i, c_j\right)^T\right] \tag{11}$$

where PST represents a series of convolution layers, BN, and activation functions, and G_e denotes the Euclidean distance function. Then, the optimized adjacency matrix A^k is computed through sparse constraint optimization, which can be represented as

$$\hat{A}^k = \text{argmin}\left\|A^k - F^e\right\|_F, \quad \text{s.t. } \left\|A^k\right\|_0 < K \tag{12}$$

where A^k represents the adjacency matrix of the kth layer; $\hat{A}^k$ represents the estimated adjacency matrix of the kth layer, obtained by minimizing the Frobenius norm between A^k and F^e; and K is a dynamic sparsity threshold that decreases with the increase of layers, focusing on the most important neighbor nodes while ignoring less significant ones.

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3) Interlayer Correlation: To further address overfitting issues caused by the limited number of samples and the complexity of GNN structures, memory-layer attention is designed. Memory-layer attention preserves critical information about the intermediate features of the model at each layer by considering the tight connections between layers, thus avoiding overfitting to the features of the current layer. In the memory-layer attention, the update of node features can be represented as follows:

$$L_k\left(C^k, W^k\right) = \mu\left(\hat{A}^k X^k \middle\| I C^k\right) W^k \tag{13}$$

where $L_k(C^k, W^k)$ represents the update of node features of the kth layer, X^k denotes the node feature matrix of the current layer, and I represents the identity matrix. Each row in the node feature matrix represents the features of a node. W^k is the weight matrix used to multiply the node features with the adjacency matrix $\hat{A}^k$. $\hat{A}^k$ is the adjacency matrix that captures the relationship between a node and its neighbors, which can be computed based on the graph's connectivity relationships and weights. μ is a nonlinear activation function used to combine and transform the results of multiplying node features with the adjacency matrix to generate new node features. In each layer's node update, node features are multiplied with an adjacency matrix and then transformed by an activation function to obtain an updated representation of the node features.

4) Global Graph Alignment: After completing node updates, the GOT distance loss is used for GGA, which comprehensively considers the structure and features of the entire graph, achieving precise GGA. The computation process is as follows:

$$L_G = \sum_{i,\hat{i},j,\hat{j}} \left[T_{i,j} \cdot f_{\cos}\left(c_i^s, c_j^t\right) + T_{\hat{i},\hat{j}} \cdot f_{\text{str}}\left(c_i^s, c_{\hat{i}}^s, c_j^t, c_{\hat{j}}^t\right)\right] \tag{14}$$

where $T_{i,j}$ and $T_{\hat{i},\hat{j}}$ are the transport matrices for node matching and node pair matching, respectively, and $f_{\cos}$ and f_{str} are two similarity computation functions.

E. Local Domain Alignment With LSL

The aim of GGA is to achieve GGA through dynamic triplet attention. However, global domain alignment may overlook subtle differences between local features, resulting in suboptimal performance when handling local details. Merely minimizing the GOT loss is not sufficient to fully alleviate the domain shift issue in FSL. Therefore, this article adopts a global–local dual alignment strategy and further designs LSL. LSL utilizes the structural connections in images for local DA, facilitating cross-domain information transfer and more comprehensively reducing domain shift.

As shown in Fig. 5, LSL consists of two parts: line message transmission and node cross-transmission.

1) Line Message Transmission: Line message transmission constructs a framework structure by connecting line segments and allows each node to transmit information by leveraging the relational information among these line segments. This enables better capture of correlations between local image regions, facilitating information transmission and alignment within the local domain. Specifically, the features of the nodes at the two endpoints are aggregated along with positional encodings of the line segments. Subsequently, these aggregated pieces of information are used to update the features of the endpoints. Such an updating mechanism allows for information to be transmitted from one node to its neighboring nodes, enhancing information exchange and complementarity between nodes. Each node representing a line segment passes its feature c_i to its neighboring nodes and receives the feature c_i of its neighboring nodes. The feature updating process is as follows:

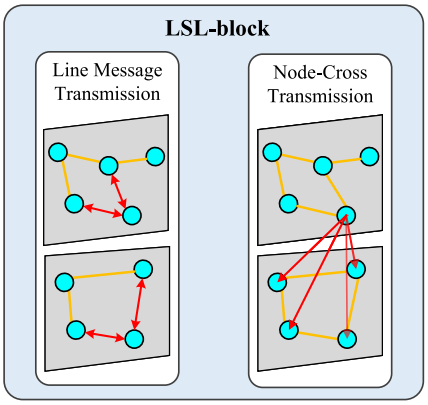

Fig. 5. Schematic of LSL. It calculates the similarity of node features between the SD and the TD through line message transmission and node cross-transmission, updating the graph structure. Ultimately, local domain alignment is achieved through similarity aggregation and computation of linear MMD distance.

$$c_i^{(m+1)} = c_i^{(m)} + \frac{\sum_{j \in N_i} \varphi_m\left(\left[c_i^{(m)} \middle\| c_j^{(m)} \middle\| c_j^e\right]\right)}{N_i} \tag{15}$$

where N_i represents the number of neighbors of node i and c_j^e denotes the positional encoding of the line segment between node i and its neighboring node j. Finally, a connectivity search across images is performed in the TD, using the learned node representations and connectivity information of the line segment graph to find nodes with similar line segment connections. This enables the model to capture local structural features through the connectivity structure effectively, facilitating local domain alignment.

2) Node Cross-Transmission: It utilizes a multihead attention mechanism to compute the feature similarity between node features from the SD and the TD and fuses it with node features to update them. It enables the network to simultaneously focus on relevant information from SD and TD, thereby identifying the corresponding or similar features between the two domains, enhancing interdomain information exchange. Specifically, first, self-attention is used to compute the feature similarity, denoted as M_{att}, between SD node features c^s and TD node features c^t, facilitating local feature matching between domains. Then, the obtained feature similarity is concatenated with node features and processed by a multilayer perceptron to obtain updated node features. The feature updating process for the mth layer is as follows:

$$c_s^{(m+1)} = c_s^{(m)} + \varphi_m\left(\left[c_s^{(m)} \middle\| M_{\text{att}}\left(c_s^{(m)}, c_t^{(m)}\right)\right]\right) \tag{16}$$

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where $c_s^{(m)}$ represents the node features in the mth round, φ_m represents a multilayer perceptron for updating features, and || denotes the vector concatenation.

3) Local Domain Alignment: LSL calculates interdomain feature similarity and minimizes differences in the feature space through similarity aggregation for local similarity alignment. The similarity aggregation process for the SD is as follows:

$$E_{\text{att}}^{s(\eta)} \leftarrow E_{\text{att}}^{s(\eta-1)} + \text{MLP}\left(\left[E_{\text{att}}^{s(\eta-1)}, M_{\text{att}}^{s(\eta-1)}\right]\right) \tag{17}$$

where $E_{\text{att}}^{(\eta)}$ represents the ηth iteration of similarity aggregation and $M_{\text{att}}^{(\eta-1)}$ denotes the $(\eta-1)$th layer of similarity information.

Finally, to ensure consistency in feature similarity between the SD and the TD, a linear MMD distance calculation is performed

$$L_M\left(E_{\text{att}}^{s(\eta)}, E_{\text{att}}^{t(\eta)}\right) = \left\| \frac{\sum_{i=1}^{C\times K+T} M\left(E_{\text{att}}^{s(\eta)}\right)}{(C\times K+T)} - \frac{\sum_{i=1}^{C\times K+T} M\left(E_{\text{att}}^{t(\eta)}\right)}{(C\times K+T)} \right\|_H^2 \tag{18}$$

where C is the number of classes, K is the number of samples per class, T is the total number of classes, and M represents the mapping function used to map the feature vectors of the output of the similarity aggregation to a specific feature space and perform the distance calculation in the feature space.

Therefore, the total loss function of GLGAT-CFSL in the training phase can be expressed as

$$L_{\text{tot}} = \mathcal{L}_{\text{fsl}}^{\text{tot}} + L_G + L_M. \tag{19}$$

IV. Experiments

In this section, we provided a detailed introduction to the HSI datasets used in the experiments and the experimental configuration. In addition, we conducted an analysis of the comparative experimental results.

A. Dataset Description

1) Source Data: The Chikusei dataset is an HSI data of the Chikusei area in Ibaraki, Japan, acquired with the Hyperspec-VNIR-CIRIS spectrometer. This dataset contains images of size 2517 × 2335 pixels with 128 bands ranging from 363 to 1018 nm. These data were captured on July 29, 2014, with the Headwall Hyperspec-VNIR-CIRIS. The spatial resolution of each pixel is 2.5 m. The dataset contains 19 different feature classes covering both urban and rural areas. Fig. 6 displays the pseudo-color composite images and the corresponding ground-truth labeled maps. In addition, Table II provides a comprehensive description of the feature classes together with the number of samples for each class.

2) Target Data: The IP dataset is a standard dataset widely used in HSI processing and feature classification studies. The dataset is acquired by the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) remote sensing sensor. It contains a 145 × 145 pixel HSI with spectral reflectance information in 224 spectral bands per pixel. The IP dataset contains 16 feature classes, such as cornfields, cotton fields, and oak forests. The ground-truth map and the corresponding color labels are shown in Fig. 7. The land cover classes and the specific quantities of pixels of each class in the IP dataset are presented in Table III.

The PU dataset is frequently utilized in the analysis of hyperspectral remote sensing images and for conducting research on feature classification. The dataset was obtained

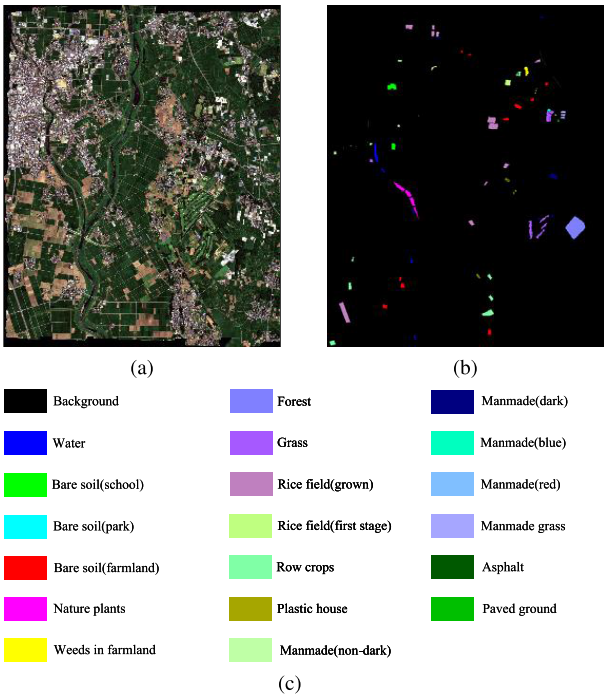

Fig. 6. Chikusei dataset. (a) Pseudo-color composite image. (b) Ground-truth image. (c) Corresponding color labels.

TABLE II
Land Cover Classes and Numbers of Samples in the Chikusei Dataset

Class	Name	Pixels
1	Water	2845
2	Bare soil(school)	2859
3	Bare soil(park)	286
4	Bare soil(farmland)	48525
5	Natural plants	4297
6	Weeds in farmland	1108
7	Forest	20516
8	Grass	6515
9	Rice field(grown)	13369
10	Rice field(first stage)	1268
11	Row crops	5961
12	Plastic house	2193
13	Manmade(non-dark)	1220
14	Manmade(dark)	7664
15	Manmade(blue)	431
16	Manmade(red)	222
17	Manmade grass	1040
18	Asphalt	801
19	Paved ground	145
Total		77592

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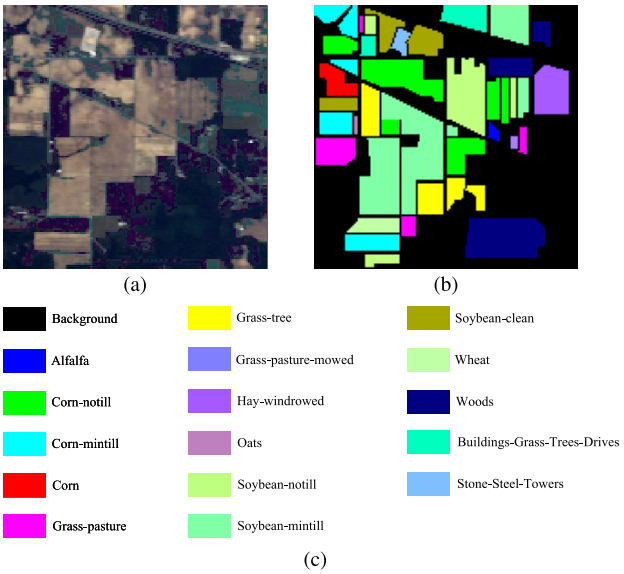

Fig. 7. Indian Pines dataset. (a) Pseudo-color image. (b) Ground-truth image. (c) Corresponding color labels.

TABLE III
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN THE INDIAN PINES DATASET

Class	Name	Pixels
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-tree	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Tress-Drives	386
16	Stone-Steel-Towers	930
Total		10249

using the AVIRIS remote sensing sensor, encompassing a high-resolution region consisting of 610 pixels in width and 340 pixels in height, with each pixel containing spectral reflectance information in 103 spectral bands. The PU dataset covers nine feature classes, such as oilseed rape fields, corn fields, and grasslands. Fig. 8 displays the ground-truth map and its related color designations. Table IV displays the land cover classes and the corresponding quantities of pixels for each class inside the PU dataset.

The SA dataset was obtained using the AVIRIS remote sensing sensor. It encompasses an area of 175 pixels in width and 512 pixels in height, located in the Salinas Valley region of California, USA. Each pixel contains spectral reflectance information in 224 spectral bands. The SA dataset covers 16 feature classes, including cornfields, vineyards, and grasslands. The ground-truth map and the corresponding color labels are shown in Fig. 9. The land cover classes and the quantities of pixels of each class in the SA dataset are presented in Table V.

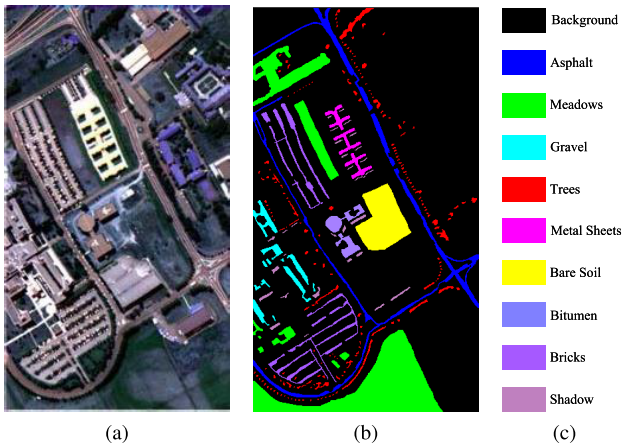

Fig. 8. Pavia University dataset. (a) Pseudo-color composite image. (b) Ground-truth image. (c) Corresponding color labels.

TABLE IV
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN THE PAVIA UNIVERSITY DATASET

Class	Name	Pixels
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Sheets	1345
6	Bare soil	5029
7	Bitumen	1330
8	Bricks	3682
9	Shadow	947
Total		42776

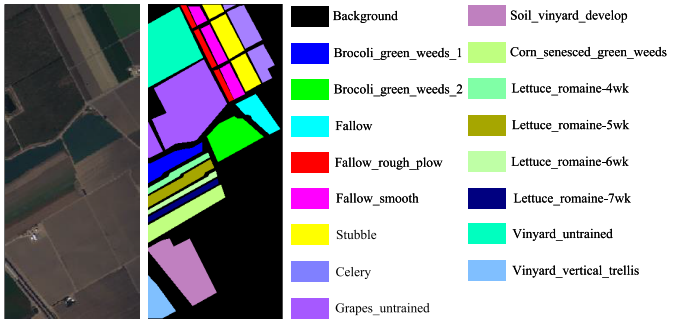

Fig. 9. Salinas dataset. (a) Pseudo-color composite image. (b) Ground-truth image. (c) Corresponding color labels.

B. Experimental Configuration

All experiments were conducted on a computer equipped with an Intel Core i7-10700KF CPU running at 3.80 GHz, 64 GB of RAM, and an RTX 4090 GPU. We utilized PyTorch as our experimental environment.

A carefully designed parameter configuration was employed to support the training and evaluation of the GLGAT-CFSL method. We utilized $9 \times 9 \times 100$ image patches as input, and the total number of training iterations was set to 10 000 to ensure sufficient learning of prototypes for each class and prevent underfitting due to insufficient training iterations. The learning rate was set to 0.001 and the weight decay was set to 0.0001 to reduce overfitting and stabilize network training.

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TABLE V
LAND COVER CLASSES AND NUMBERS OF SAMPLES IN THE SALINAS DATASET

Class	Name	Pixels
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
Total		54129

Regarding hyperparameters, during the episodic training phase, the number of samples per class in the support set was set to 1, and the number of samples per class in the query set was set to 19 to evaluate the model's generalization ability on unseen data.

Three metrics were employed to evaluate the experimental results: average accuracy (AA), overall accuracy (OA), and kappa coefficient (KC).

C. Ablation Experiment

An ablation analysis was conducted to assess the contributions of three key modules, FSL, GGA, and LSL, in the GLGAT-CFSL framework. In the ablation analysis, we progressively added the proposed global and local domain alignment modules (GGA and LSL) to the FSL module, resulting in four variants: "Only FSL," "FSL+GGA," "FSL+LSL," and "FSL+GGA+LSL (GLGAT-CFSL)," to explore their impact on overall performance. All ablation experiments were conducted using five labeled samples for training. The results of the ablation experiments are presented in Tables VI–VIII.

The experimental results present in Tables VI–VIII indicate that, taking the IP dataset as an example, the "Only FSL" achieves an OA of 70.29% on the IP dataset. Compared to "Only FSL," the variants "FSL+GGA" and "FSL+LSL" separately added the DA modules GGA and LSL, with the addition of DA modules, and the network's classification performance improved by 3%−3.7%. It suggests that combining DA methods with FSL enables the model to better adapt to different domains' data distributions, thereby improving the model's performance on TDs. Furthermore, "FSL+GGA+LSL (GLGAT-CFSL)" exhibited a further improvement in network classification performance by 5% compared to "FSL+LSL" and "FSL+GGA." This suggests that the global-to-local DA can more comprehensively mitigate the negative effects of domain shift.

D. Results of Comparative Experiments

A comprehensive comparison was conducted between our proposed GLGAT-CFSL method and various state-of-the-art algorithms, including traditional SVM, CNN-based 3-D-CNN, FSL method DFSL, as well as five advanced DA-based FSL methods: DCFSL, Gia-CFSL, CMFSL, ADAFSL, and CFSL-KT. All comparative experiments were conducted using five labeled samples for training. Nondomain-adaptation methods, such as SVM, 3-D-CNN, and DFSL, were directly trained and tested on the IP, PU, and SA HSI datasets. On the other hand, DCFSL, Gia-CFSL, CMFSL, ADAFSL, CFSL-KT, and our method utilized the Chikusei dataset as the SD and the IP, PU, and SA datasets as the TDs.

The results of the comparative experiments are shown in Tables IX–XI. Traditional method SVM performed the worst, indicating that traditional machine learning methods cannot address the issue of small-sample problems. CNN-based 3-D-CNN outperformed traditional SVM, yet still failed to address the problem of limited labeled samples. The FSL method DFSL, leveraging meta-learning principles, conducted effective knowledge inference and generalization from a small number of samples. Domain-adaptation-based FSL methods, such as DCFSL, Gia-CFSL, CMFSL, ADAFSL, and CFSL-KT, further improved classification performance by introducing DA modules. Our proposed GLGAT-CFSL, also a domain-adaptation-based FSL method, achieved the best classification results by jointly aligning global and local domains more comprehensively. Compared to other state-of-the-art domain-adaptation-based FSL methods, GLGAT-CFSL achieved an improvement in OA by 4.5%–14% on the IP dataset, 1.2%–7.5% on the PU dataset, and 1.2%–5.1% on the SA dataset.

Figs. 10–12 visualize the ground-truth maps of different methods on the three HSI datasets. As shown in Figs. 10–12, it can be inferred that, compared with other methods, GLGAT-CFSL classifies more samples correctly and provides the most accurate classification structure map. This indicates that GLGAT-CFSL has a significant advantage in the field of small-sample HSIC.

What is more, we employed t-distributed stochastic neighbor embedding (t-SNE) to visualize the embeddings in Fig. 13, presenting the 2-D projection features of ADAFSL, CFSL-KT, and GLGAT-CFSL across the three TD datasets to enhance the interpretability of our proposed models. In Fig. 13, our GLGAT-CFSL consistently outperforms the other two algorithms. Notably, GLGAT-CFSL demonstrates clear distinctions among the second class "Corn_notill," the third class "Corn_mintill," and the fourth class "Corn." On the PU dataset, the first class "Bitumen" and the seventh class "Asphalt" share similar spectral features, making them challenging to distinguish. Compared to ADAFSL and CFSL-KT, GLGAT-CFSL exhibits better differentiation between these two classes. On the SA dataset, distinguishing the eighth class "Grapes_untrained" and the 15th class "Vinyard_untrained" is challenging. While ADAFSL and CFSL-KT tend to group these two classes into one cluster, GLGAT-CFSL provides better classification.

V. DISCUSSION

A. Influence of Hyperparameter

This section examines the influence of various parameter configurations on the performance of the GLGAT-CFSL.

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TABLE VI
ABLATION EXPERIMENTS THROUGH COMBINATIONS OF MODULES ON THE INDIAN PINES

Dataset	Indian Pines			
Model	Only FSL	FSL+LSL	FSL+GGA	FSL+GGA+LSL (GLGAT-CFSL)
OA	70.29±1.00	73.36±0.81	73.98±1.10	**78.50±1.59**
AA	79.13±1.00	81.22±0.75	81.67±1.02	**83.96±1.17**
KC	66.59±1.09	68.95±0.89	69.45±1.23	**79.98±1.73**

TABLE VII
ABLATION EXPERIMENTS THROUGH COMBINATIONS OF MODULES ON PAVIA UNIVERSITY

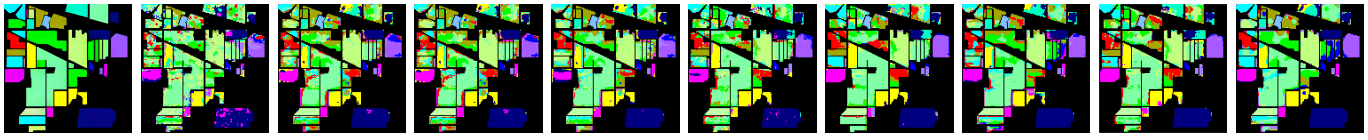
Dataset	Pavia University			
Model	Only FSL	FSL+LSL	FSL+GGA	FSL+GGA+LSL (GLGAT-CFSL)
OA	81.65±1.20	84.97±1.09	85.87±1.76	**88.17±1.06**
AA	82.16±1.20	85.03±1.61	86.60±1.89	**93.71±0.53**
KC	76.26±1.53	79.55±1.37	80.03±2.42	**88.35±1.17**

TABLE VIII
ABLATION EXPERIMENTS THROUGH COMBINATIONS OF MODULES ON THE SALINAS

Dataset	Salinas			
Model	Only FSL	FSL+LSL	FSL+GGA	FSL+GGA+LSL (GLGAT-CFSL)
OA	87.58±0.37	91.09±0.33	91.67±0.40	**92.28±0.71**
AA	91.04±0.47	93.55±0.34	94.83±0.41	**96.65±0.43**
KC	86.78±0.40	90.08±0.37	90.42±0.44	**95.85±0.79**

TABLE IX
CLASSIFICATION ACCURACIES FOR DIFFERENT METHODS ON TARGET SCENES AND OVERALL CLASSIFICATION ACCURACY (%)—INDIAN PINES DATASET (FIVE LABELED SAMPLES FROM TD)

Class	Classification algorithms								
	SVM	3-D-CNN	DFSL[22]	DCFSL[27]	Gia-CFSL[34]	CMFSL[31]	ADAFSL[32]	CFSL-KT[33]	GLGAT-CFSL
1	41.35	93.12	92.68	97.56	95.61	93.53	**100**	98.12	**100**
2	42.81	37.44	26.84	22.28	38.67	48.32	50.62	44.31	**78.83**
3	39.04	24.85	37.45	52.97	44.38	**82.23**	63.42	69.95	81.93
4	51.49	34.24	71.98	88.79	85.13	90.30	80.82	87.58	**95.79**
5	59.33	81.23	61.72	77.41	78.45	92.04	78.46	73.83	**96.99**
6	65.72	71.95	90.14	94.62	73.81	**96.98**	90.93	94.85	95.03
7	80.83	**100**	97.65	**100**	95.65	99.53	99.85	**100**	99.74
8	71.23	52.84	81.64	93.62	80.38	94.35	91.45	92.86	**98.91**
9	81.52	87.43	99.42	96.62	98.67	99.58	**100**	99.64	99.67
10	33.47	44.84	58.53	63.39	61.12	64.96	58.85	64.94	**79.82**
11	32.14	58.04	60.78	59.72	70.65	37.68	68.42	60.92	**80.96**
12	34.96	33.64	30.78	40.89	53.86	49.46	58.52	60.90	**70.03**
13	60.89	91.21	97.50	98.39	**99.60**	96.35	97.20	99.15	98.75
14	81.21	52.97	89.52	94.52	92.29	84.46	85.62	90.80	**97.84**
15	21.14	56.42	44.88	56.69	67.95	58.45	**69.48**	68.20	68.93
16	88.19	89.52	96.59	96.73	99.43	98.37	97.46	99.58	**100**
OA	43.58	55.84	59.96	64.40	67.04	66.89	70.24	74.96	**78.50**
AA	58.68	55.84	59.96	64.40	67.04	66.89	79.82	81.48	**82.96**
KC	37.21	49.95	54.73	66.67	69.23	66.96	68.42	73.26	**79.98**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 10. Indian Pines. (a) Ground-truth map. Classification results for different methods. (b) SVM. (c) 3-D-CNN. (d) DFSL. (e) DCFSL. (f) Gia-CFSL. (g) CMFSL. (h) ADAFSL. (i) CFSL-KT. (j) Our method.

Specifically, we focus on two crucial hyperparameters: the embedded feature dimension and the base learning rate setting. For these hyperparameters, we use the tenfold cross-validation method to determine the optimal settings. Through in-depth

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TABLE X

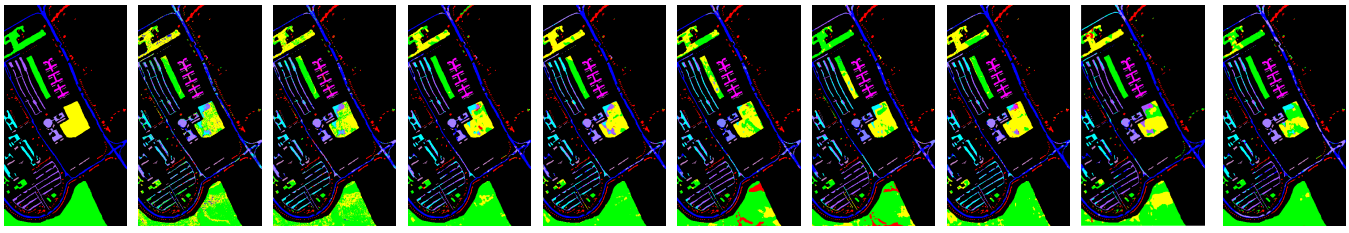
CLASSIFICATION ACCURACIES FOR DIFFERENT METHODS ON TARGET SCENES AND OVERALL CLASSIFICATION ACCURACY (%)—PAVIA UNIVERSITY DATASET (FIVE LABELED SAMPLES FROM TD)

Class	Classification algorithms								
	SVM	3-D-CNN	DFSL[22]	DCFSL[27]	Gia-CFSL[34]	CMFSL[31]	ADAFSL[32]	CFSL-KT[33]	GLGAT-CFSL
1	77.91	54.21	68.66	70.22	86.77	74.81	85.42	**96.84**	95.85
2	73.19	69.50	81.66	87.01	89.18	92.11	85.94	86.06	**96.95**
3	46.89	66.17	53.01	46.23	74.78	79.58	58.68	79.54	**85.09**
4	62.32	78.13	87.38	92.58	90.75	86.58	85.42	**92.96**	91.92
5	93.41	92.71	96.49	98.93	97.11	99.68	**99.75**	97.82	98.80
6	35.28	63.24	72.25	87.92	74.72	75.83	75.16	**94.81**	92.57
7	45.21	79.44	73.49	79.62	83.22	91.63	85.20	90.16	**92.28**
8	70.71	57.72	58.72	59.07	52,69	73.48	80.18	72.72	**90.95**
9	98.35	97.47	97.47	98.18	98.90	98.25	98.92	93.62	**98.98**
OA	62.24	64.47	78.35	80.66	83.82	85.28	85.46	86.94	**88.17**
AA	67.32	70.28	80.21	80.84	82.18	83.98	83.64	90.60	**91.71**
KC	54.94	56.73	70.49	74.33	78.30	80.49	81.40	85.42	**88.35**

TABLE XI

CLASSIFICATION ACCURACIES FOR DIFFERENT METHODS ON TARGET SCENES AND OVERALL CLASSIFICATION ACCURACY (%)—SALINAS DATASET (FIVE LABELED SAMPLES FROM TD)

Class	Classification algorithms								
	SVM	3-D-CNN	DFSL[22]	DCFSL[27]	Gia-CFSL[34]	CMFSL[31]	ADAFSL[32]	CFSL-KT[33]	GLGAT-CFSL
1	97.75	95.19	97.12	94.25	95.40	98.27	99.58	**99.84**	99.04
2	87.32	96.29	96.21	99.17	98.52	99.38	99.49	98.83	**99.51**
3	83.45	91.54	51.39	62.27	94.58	97.72	94.74	97.89	**99.61**
4	95.11	96.39	98.84	98.27	99.25	99.03	99.06	98.96	**99.40**
5	91.98	88.82	88.92	97.88	91.25	94.69	87.47	**98.28**	96.30
6	97.63	98.14	98.95	98.78	98.92	98.92	96.85	99.12	**99.44**
7	93.96	97.94	97.29	98.29	99.38	98.83	99.58	99.59	**99.66**
8	60.95	66.04	**90.96**	89.12	71.76	68.66	78.72	64.61	88.64
9	96.03	95.51	84.44	86.63	97.71	98.68	98.71	99.39	**99.84**
10	70.34	68.29	70.05	77.17	68.48	88.21	80.04	**92.82**	90.93
11	88.72	91.59	49.51	98.98	97.98	**99.44**	98.87	97.94	97.17
12	96.92	98.88	93.23	71.39	98.02	99.44	99.50	94.67	**99.74**
13	97.38	93.36	98.65	98.89	98.50	98.35	99.24	99.46	**99.57**
14	91.49	94.62	98.86	73.03	99.05	99.09	99.22	99.76	**99.84**
15	61.95	69.23	56.94	63.99	86.03	80.09	77.53	85.42	**86.36**
16	85.42	81.11	83.86	73.48	88.61	**95.22**	85.29	94.85	90.40
OA	80.37	82.62	85.65	87.21	89.36	89.52	90.42	91.10	**92.28**
AA	85.12	86.40	90.58	93.24	93.56	94.02	94.26	95.24	**95.85**
KC	78.17	81.76	84.93	88.14	88.26	88.54	90.20	93.48	**95.24**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 11. Pavia University. (a) Ground-truth map. Classification results for different methods. (b) SVM. (c) 3-D-CNN. (d) DFSL. (e) DCFSL. (f) Gia-CFSL. (g) CMFSL. (h) ADAFSL. (i) CFSL-KT. (j) Our method.

study and tuning of these key hyperparameters, we successfully improve the performance and generalization ability of the GLGAT-CFSL model, enabling it to perform better in the cross-domain FSL task. This exploration of parameter tuning provides solid experimental support and theoretical guidance and is a vital reference for model optimization and future research.

1) Embedding Feature Dimension: The embedded feature dimensions of different settings have been extensively studied, including 32, 64, 128, and 256. The experimental results

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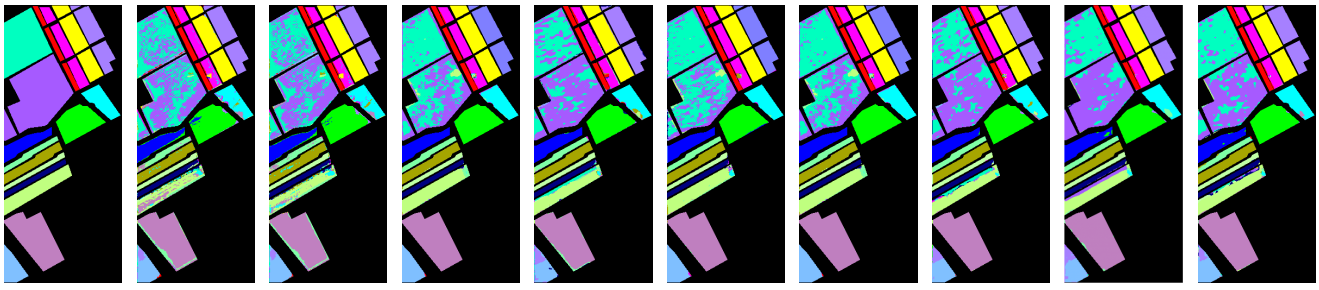

Fig. 12. Salinas. (a) Ground-truth map. Classification results for different methods. (b) SVM. (c) 3-D-CNN. (d) DFSL. (e) DCFSL. (f) Gia-CFSL. (g) CMFSL. (h) ADAFSL. (i) CFSL-KT. (j) Our method.

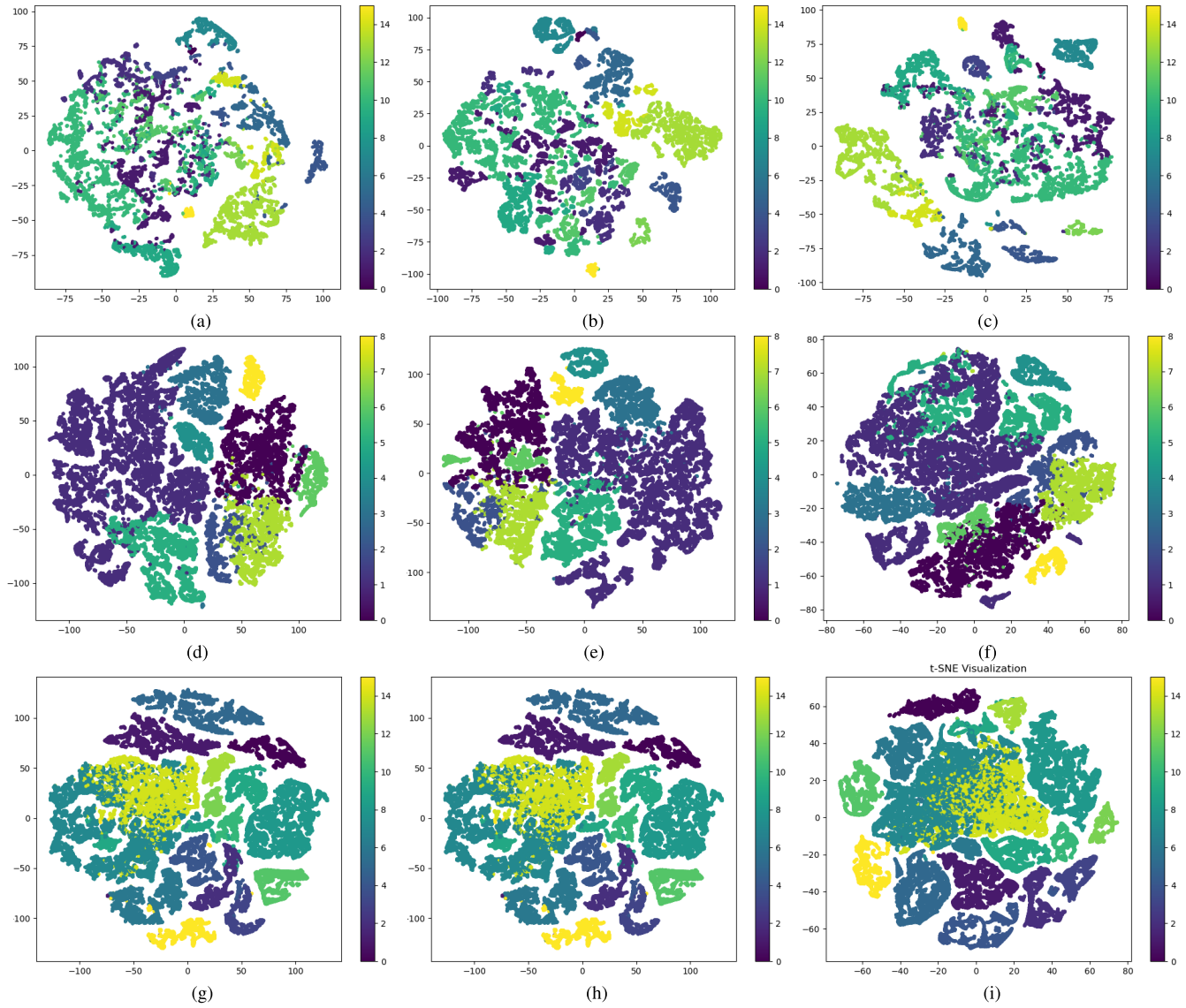

Fig. 13. 2-D feature visualization on three datasets. (a) ADAFSL features on IP. (b) CFSL-KT features on IP. (c) GLGAT-CFSL features on IP. (d) ADAFSL features on PU. (e) CFSL-KT features on PU. (f) GLGAT-CFSL features on PU. (g) ADAFSL features on SA. (h) CFSL-KT features on SA. (i) GLGAT-CFSL features on SA.

under different embedding feature dimensions are shown in Table XII. The experimental results reveal a strong relationship between embedding feature dimensions and model performance. In this parameter tuning, we find that the 64-D embedding feature dimension achieves the best performance on our dataset. Taking the IP dataset as an example, compared to the other three dimensions, the 64-D embedding feature dimension improved OA by approximately 0.1%–1.3%. Specifically, the 64-D embedding feature dimension not only performs well on the training data but also shows better

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TABLE XII
EFFECT OF DIFFERENT EMBEDDING FEATURE DIMENSIONS ON MODEL PERFORMANCE

Dataset	IP				PU				SA			
d emb	32	64	128	256	32	64	128	256	32	64	128	256
OA	78.42	**78.50**	78.09	77.15	86.84	**88.17**	87.43	86.92	90.45	**92.28**	91.24	90.66
AA	82.56	**83.96**	83.65	82.91	92.06	**93.71**	93.24	92.25	94.48	**96.65**	95.52	94.52
KC	78.06	**79.98**	79.25	77.53	87.15	**88.35**	87.85	86.69	94.46	**95.85**	94.98	93.62

TABLE XIII
INFLUENCE OF DIFFERENT LEARNING RATES ON MODEL PERFORMANCE

Target Source	1e-5	1e-4	1e-3	1e-2	1e-1
IP(OA)	74.61	76.27	**78.50**	76.89	74.20
PU(OA)	84.75	86.87	**88.17**	85.54	84.05
SA(OA)	91.06	91.18	**92.28**	90.69	90.03

generalization ability on the unseen test data, enabling the model to perform the classification task more accurately.

2) Learning Rate: Smaller learning rates may result in slow convergence during the training process, while larger learning rates may induce model oscillations. Therefore, the base learning rate has been thoroughly investigated to find the optimal setting. We explored the influence of different learning rate settings on model training, including $1e^{-5}$, $1e^{-4}$, $1e^{-3}$, $1e^{-2}$, and $1e^{-1}$. The experimental results at different learning rates are shown in Table XIII. From the experimental results, taking the IP dataset as an example, the learning rate of $1e^{-3}$ improved OA by approximately 1.6%–4.3% compared to other settings. Therefore, we chose $1e^{-3}$ as the optimal base learning rate. This setting ensures fast convergence during training while maintaining the stability of the model, resulting in better performance.

B. Evaluation on the TDCNN

As described in Section III, TDCNN achieves cross-dimensional interaction between spectral and spatial dimensions by combining dynamic convolution and TA, thereby enhancing the network's ability to extract crucial features during training. Therefore, in this section, we primarily analyze the influence of TDCNN components on network performance.

First, in order to demonstrate the effectiveness of the dynamic convolution and TA modules in TDCNN, we conducted ablation experiments on them. Specifically, while keeping other experimental settings constant, we sequentially added the dynamic convolution module and the TA module to a deep residual 3-D CNN network. Specifically, we designed three variants, namely, "ResNet+3-D-CNN (Res+3DCNN)," "ResNet+Dynamic Conv (Res+dc)," and "ResNet+Triaxial Attention (Res+TA)," and compared them with our method (Res+dc+TA). Keeping other experimental settings unchanged, we explored the influence of TDCNN on network performance. The experimental results are shown in Fig. 14. The results indicate that, taking the IP dataset as an example, the OA of the deep residual 3-D CNN is 76.5%. Upon adding dynamic convolution or TA, the OA of "Res-DC" and "Res+TA" increases by approximately 0.5%–1.3%. In addition, our method "Res+dc+TA" achieves an OA improvement of approximately 2%, demonstrating the optimal performance. This suggests that TDCNN enhances feature expression capability and cross-dimensional information interaction by combining dynamic convolution and TA mechanisms, further improving deep feature extraction capabilities.

Second, we compared the performance of TDCNN with other state-of-the-art networks (CPCNN, DFINET, and ACL-CNN) in the cross-domain HSIC task. Specifically, while keeping other settings constant, we replaced TDCNN with CPCNN, DFINET, and ACL-CNN, resulting in four variants: "GLGAT-CPCNN," "GLGAT-DFINET," and "GLGAT-ACLCNN," respectively. We then compared these variants with our method "GLGAT-TDCNN." The experimental results are shown in Fig. 15. The results indicate that, taking the IP dataset as an example, compared to other state-of-the-art networks, the OA of GLGAT-CFSL using TDCNN as the feature extractor improves by approximately 0.3%–1.5%. It suggests that TDCNN can establish more consistent and comprehensive feature representations between the SD and the TD, further promoting domain alignment.

C. Different Numbers of Training Samples

We focused on the number of FSL training samples to explore the effect of sample size on model performance to validate the effectiveness of the proposed GLGAT-CFSL further. Specifically, different settings were tried from 1 to 5 samples to examine the performance of the model with different numbers of samples. The experimental results are shown in Fig. 16, which show that appropriately increasing the number of FSL training samples helps improve the model's classification accuracy and generalization performance. When the number of samples is too small, the model may face the problem of overfitting. As the increase in training samples for the TD, there is a notable improvement in the classification accuracy of the TD, even when only limited supervised information is available for each class. GLGAT-CFSL can fully utilize them to learn a prototype representation for each class. After comprehensive tradeoffs, we chose five samples per class as the optimal number of FSL training samples. This setting can adequately capture class features while avoiding the risk of overfitting. Most notably, the GLGAT-CFSL method consistently secures the top position, providing strong evidence for the effectiveness of the proposed networks.

D. Computational Complexity

To assess the computational efficiency of different methods, we analyzed their training time, testing time, number of

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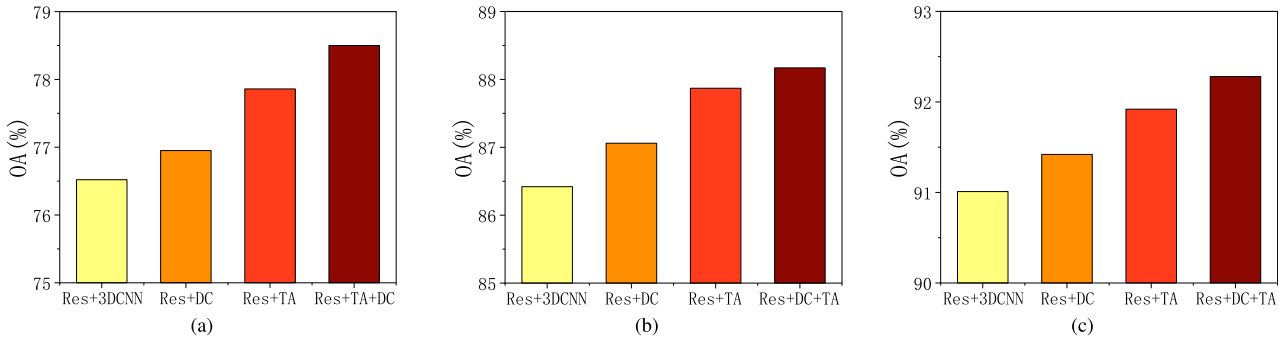

Fig. 14. Classification performance of the Res+3DCNN, Res+DC, Res+TA, and out method Res+DC+TA on the three datasets. (a) Indian Pines. (b) Pavia University. (c) Salinas.

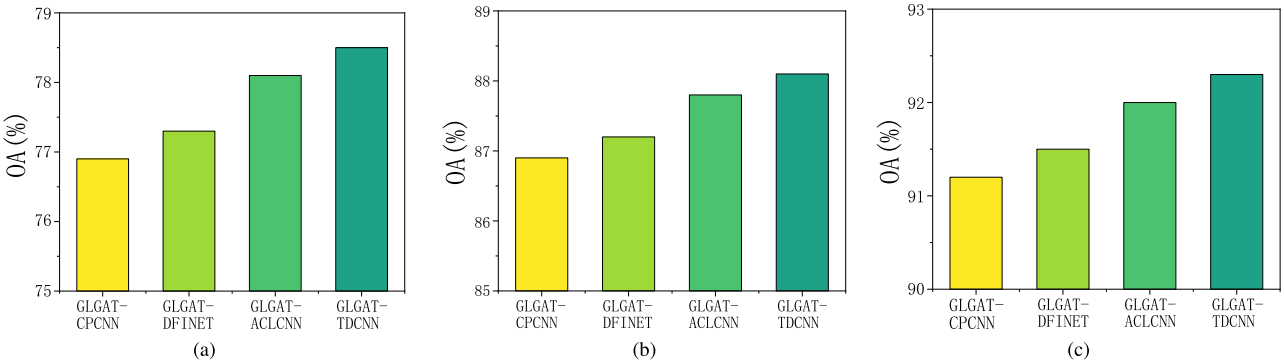

Fig. 15. Classification performance of the GLGAT-CPCNN, GLGAT-DFINET, GLGAT-ACLCNN, and our method GLGAT-TDCNN on the three datasets. (a) Indian Pines. (b) Pavia University. (c) Salinas.

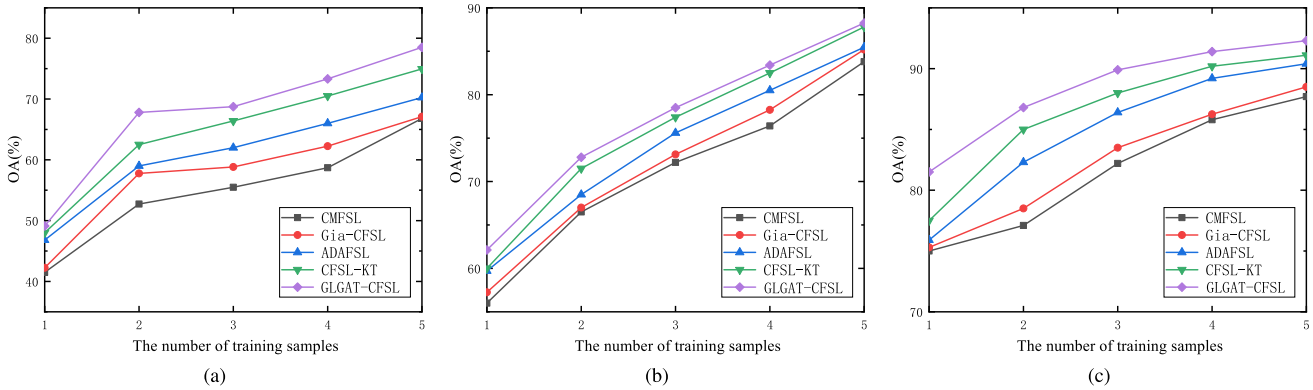

Fig. 16. Effect of different numbers of training samples on the performance of different methods on three TD datasets. (a) Indian Pines. (b) Pavia University. (c) Salinas.

TABLE XIV
TRAINING TIME, TESTING TIME, FLOPS, AND PARAMETERS ON EACH DATASET WITH DIFFERENT METHODS

	Methods	CMFSL	ADAFSL	CFSL-KT	Gia-CFSL	GLGAT-CFSL
Indian Pines	Training time	1886.42s	3990.12s	2014.82s	3024.97s	2834.62s
	Testing time	1.42s	2.80s	1.45s	1.05s	1.08s
	FLOPS	0.2441G	0.0424G	0.5978G	5.3325G	5.3236G
	Parameters	0.6139M	0.0242M	0.4482M	0.1851M	0.1824M
PaviaU	Training time	982.50s	1575.24s	2034.27s	2034.27s	1942.04s
	Testing time	5.26s	9.01s	6.98s	4.08s	4.05s
	FLOPS	0.2441G	0.0424G	0.5978G	5.3325G	5.3236G
	Parameters	0.6139M	0.0242M	0.4482M	0.1851M	0.1824M
Salina	Training time	1692.78s	2445.06s	2605.06s	5359.06s	5196.35s
	Testing time	7.24s	13.06s	9.87s	5.87s	5.46s
	FLOPS	0.2441G	0.0424G	0.5978G	5.3325G	5.3236G
	Parameters	0.6139M	0.0242M	0.4482M	0.1851M	0.1824M

parameters, and floating-point operations (FLOPs) to evaluate the computational complexity of deep learning approaches, as shown in Table XIV. In these experiments, we utilized a TD dataset with five labeled samples per class. Training

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time is primarily spent on transfer learning, while testing time encompasses the prediction of unlabeled samples in each HSI dataset. From Table XIV, it can be observed that GLGAT-CFSL exhibits higher training time and FLOPs, attributed to the increased complexity of graph-based models. However, GLGAT-CFSL achieves the highest classification accuracy.

VI. Conclusion

In this study, we propose the GLGAT-CFSL framework to address the challenges of DA in HSIC. The proposed framework combines global–local domain alignment and cross-dimensional interaction feature extractor to enhance classification performance in cross-domain scenarios with limited labeled data. Our research findings demonstrate that GLGAT-CFSL effectively overcomes the limitations of traditional DA methods. By integrating both global and local domain alignment, the GLGAT-CFSL framework achieves improved DA and enhanced classification performance. Comparing the performance of GLGAT-CFSL with existing methods such as SVM, 3-D-CNN, DFSL, DCFSL, Gia-CFSL, CMFSL, ADAFSL, and CFSL-KT, consistent improvements in classification results were observed across three different TDs.

Looking forward, our research will focus on developing lightweight domain alignment strategies based on GNNs. This approach will significantly reduce computational and memory overhead, thereby enhancing the efficiency and applicability of GNN methods in cross-domain tasks. In addition, we will explore and introduce more effective attention mechanisms, which will further improve the performance of GNNs, making feature extraction and representation more accurate and efficient. By advancing these research directions, we aim to overcome the current limitations of GNN methods in handling large-scale graph processing, thereby enhancing their practicality and scalability in HSIC and other complex graph learning tasks.

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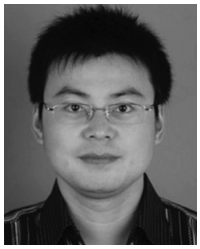

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